

CO5430 - Image Processing

Project Proposal and Plan

Group No. 15

7th July, 2026

Problem Statement and Motivation

Melanoma is one of the most aggressive forms of skin cancer, but survival rates improve dramatically when it is detected early. Diagnosis is typically performed visually by dermatologists using dermoscopic imaging. It's a process that is subjective, dependent on clinician experience, and not always accessible in resource-limited settings. This project addresses melanoma detection as a computer vision problem, specifically a binary image classification task in which a dermoscopic image of a skin lesion is classified as either benign or malignant. The objective is to build and evaluate an automated classifier that could support, rather than replace, clinical screening by flagging lesions that warrant closer examination. Also, this project gives the group direct experience with the full computer vision workflow applied to a real medical imaging dataset.

Related Work

Prior work on melanoma classification has largely converged on convolutional neural network approaches. The International Skin Imaging Collaboration challenge series has also played a significant role in standardizing dermoscopic image classification as a benchmark problem, establishing sensitivity. This project situates itself within this body of work by implementing and comparing a simple baseline against a fine-tuned deep model on a specific Kaggle-sourced dermoscopic dataset, with an focus on evaluation accuracy and explainability rather than creative architecture design.

Dataset

The project will use the **Melanoma Cancer Image Dataset** from Kaggle, which consists of dermoscopic images of skin lesions labeled as benign or malignant, sized uniformly at 224 by 224 pixels in RGB. The dataset contains on the order of ten thousand images. The existing train and test partitions provided with the dataset will be used where suitable, with a portion of the training set held out as a validation set. Since the dataset consists of publicly released dermoscopic images rather than patient records, privacy risk is minimal. Limitations of

the dataset (class imbalance, image quality variation, lack of demographic diversity) will be discussed explicitly rather than assumed away.

Planned Method

A simple baseline will first be established (a compact CNN trained from scratch or a pre-trained backbone used only as a frozen feature extractor with a linear classification head) to provide a lower bound against which improvements can be measured. The primary method will be a fine-tuned pretrained convolutional backbone, adapted to the binary classification task. Data augmentation, including random flips, rotations, and color jitter, will be applied during training to improve generalization given the relatively limited size of the dataset for deep fine-tuning.

Evaluation Plan

Model performance will be reported using accuracy, precision, sensitivity (F1-score, and a confusion matrix) with particular attention to sensitivity given that a false negative in this context corresponds to a missed cancer diagnosis. Also, a saliency visualization technique will be used to illustrate which regions of a lesion image the model relies on for its prediction. The minimum goal is a working baseline classifier alongside a fine-tuned deep model. The expected goal extends this with a full metric suite, at least one ablation or comparison study, and a qualitative analysis of failure cases.

Timeline

Group formation and dataset access will be finalized in the first week of July and this proposal is submitted by 14 July. By the checkpoint on 28 July, the preprocessing pipeline and baseline model will be implemented and producing initial results. By the checkpoint on 18 August, the fine-tuned model will be functional with preliminary quantitative and qualitative results available. Experiments will be frozen by 25 August, after which effort will shift to analysis, ablation completion, and report writing, with a draft report ready by 1 September ahead of final submission and demonstration on 7 September.

Risks and Mitigations

The most significant risk is class imbalance between benign and malignant samples, which will be mitigated through class-weighted loss functions or targeted augmentation of the minority class. A second risk is overfitting given the moderate dataset size relative to the capacity of modern CNN backbones, which will be addressed through transfer learning, data augmentation, and early stopping based on validation performance rather than training from scratch. A third risk is that the model may learn spurious correlations from artifacts present in some dermoscopic images, such as ruler markings, ink markings, or hair.